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"Generating matrix for Generalized Fibonacci numbers and Fibonacci polynomials

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Abstract

Many researchers have been working on recurrence relation sequences of numbers and polynomials which are useful topic not only in mathematics but also in physics, economics and various applications in many other fields. There are many useful identities on recurrence relation sequence but there main problem to find any term of recurrence relation sequence we need to find previous all terms of recurrence relation sequence of numbers and polynomials. There were many important theorems obtained on recurrence relation sequences. In this paper we have given special identity for generalized Fibonacci sequence of number and Fibonacci sequence of polynomials. These identities are very useful to represent Fibonacci generalized sequence of numbers and Fibonacci sequence of polynomials in the form of matrix. Authors define a special formula in this paper by this we can find special representation of Fibonacci generalized sequence of numbers and Fibonacci sequence of polynomials in the form of matrix. So, we can say that this paper is generalization of property of Fibonacci sequence of number and Fibonacci sequence of polynomials

Keywords: Generalized, recurrence relation, sequence, matrix.

1. Introduction

1.1 Fibonacci numbers

Italian Mathematician Leonardo of Pisa who is also known as by his nickname Fibonacci (1170-1240) he wrote (Book of the Abacus) in 1202. He was 1st European mathematician which work on Indian and Arabian mathematics. He gave a special type sequence [5, 6, 9]

$$F_n = F_{n-1} + F_{n-2} \quad ,n \geq 2 \tag{1}$$

With initial Term $F_0 = 0$ and $F_1 = 1$

Edouard Lucas dominated the field recursive series during the period 1878-1891 he was 1st mathematician who applied Fibonacci's name for sequence (1) and it has been known as Fibonacci sequence since then. Lucas sequence given by [10, 11, 12].

$$L_n = L_{n-1} + L_{n-2} n \geq 2 \tag{2}$$

With initial term, $L_0 = 2$ $L_1 = 1$

Terms of the Lucas sequence are called Lucas numbers.

1.2 Generalized Fibonacci sequences of numbers

Generalized Fibonacci sequence is defined as [8, 9].

$$F_k = aF_{k-1} + bF_{k-2}, k \geq 2 \; with \; F_0 = p, F_1 = q \tag{3}$$

where p, q, a & b are positive integers

1.3 Consider a particular case of generalized Fibonacci sequence

by equation (1.2.1) as

$$V_k = aV_{k-1} + bV_{k-2}, k \geq 2 \; with \; V_0 = 0, V_1 = 1 \tag{4}$$

Where a & b are positive integers.

1.4 Fibonacci sequences of polynomials

Fibonacci polynomial $F_n(x)$ studied by E.C. Catalan is define by

$$F_{n+2}(x) = xF_{n+1}(x) + F_n(x)$$

Where $n = 0,1,2,....$ and $F_0(x) = 0$, $F_1(x) = 1$.

1.5 Generalized Fibonacci sequences of polynomials

We defined a recurrence relation sequence of polynomials $G_n(x)$ named as Generalized Fibonacci polynomial defined is given by

$$G_{n+2}(x) = axG_{n+1}(x) + bG_n(x) \tag{5}$$

Where $n = 0,1,2,....$ and $G_0(x) = 0$, $G_1(x) = 1$.

1.6 Sequence of tri-diagonal matrices some entries are variable

We defined a special sequence of tridiagonal matrix {A(n)} such that

$$[g_{i,j}] = \left\{ \begin{array}{c} g_{i,j} = ax \quad if \;\; i = j \\ g_{i,j} = 1 \;\; if \; i = j+1 \\ g_{i,j} = -b \;\; if \;\; i = j-1 \\ g_{i,j} = 0 \;\; if \; oterwise \end{array} \right\} \tag{6}$$

$$\boldsymbol{A(n)} = \begin{bmatrix} \boldsymbol{ax} & \boldsymbol{-b} & \boldsymbol{0} & \cdots & \cdots & \boldsymbol{0} \\ \boldsymbol{1} & \boldsymbol{ax} & \boldsymbol{-b} & \cdots & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{1} & \boldsymbol{ax} & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \boldsymbol{ax} & \boldsymbol{-b} \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \cdots & \boldsymbol{1} & \boldsymbol{ax} \end{bmatrix}$$

Then determinant of $\boldsymbol{A(n)}$ are

$$|A(1)| = g_{1,1}$$

$$|A(2)| = g_{1,1}g_{2,2} - g_{2,1}g_{1,2}$$

$$\cdots$$

$$|A(n)| = g_{n,n}|A(n-1)| - g_{n,n-1}g_{n-1,n}|A(n-2)| \tag{7}$$

1.7 Sequence of tri-diagonal matrices all entries are numbers

We defined a special sequence of tri-diagonal matrix $\{D(n)\}$ such that

$$[g_{i,j}] = \left\{ \begin{array}{c} g_{i,j} = a \quad if \;\; i = j \\ g_{i,j} = 1 \;\; if \; i = j+1 \\ g_{i,j} = -b \;\; if \;\; i = j-1 \\ g_{i,j} = 0 \;\; if \; oterwise \end{array} \right\} \tag{8}$$

$$\boldsymbol{D(n)} = \begin{bmatrix} \boldsymbol{a} & \boldsymbol{-b} & \boldsymbol{0} & \cdots & \cdots & \boldsymbol{0} \\ \boldsymbol{1} & \boldsymbol{a} & \boldsymbol{-b} & \cdots & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{1} & \boldsymbol{a} & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \boldsymbol{a} & \boldsymbol{-b} \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \cdots & \boldsymbol{1} & \boldsymbol{a} \end{bmatrix}$$

Then determinant of $\boldsymbol{D(n)}$ are of the type

$$|D(1)| = g_{1,1}$$

$$|D(2)| = g_{1,1}g_{2,2} - g_{2,1}g_{1,2}$$

$$|D(n)| = g_{n,n}|D(n-1)| - g_{n,n-1}g_{n-1,n}|D(n-2)| \tag{9}$$

Theorem 1: - $|A(n)| = G_{k+1}\ for\ all\ n \geq 1$, where $|A(n)|$the determinant of nth term of above is define (6) sequence of matrix and G_{k+1} is $(k+1)th$ term of generalized Fibonacci sequence of polynomials (5)

Proof: - We will prove this result by principle mathematical induction

For $n = 1$

$|A(1)| = ax$ also $G_2 = ax$

So, we can say $|A(1)| = G_2$

Which is true for $n = 1$

Consider result is true for $n \leq k$ (our hypothesis)

So we have $|A(n)| = G_{k+1}$

Now we will show that result is also true for $n = k + 1$

Consider $|A(k+1)| = g_{k+1,k+1}|A(k)| - g_{k+1,k}g_{k,k+1}|A(k-1)|$ $\quad by\ (6)$

$By\ (6)$ $\quad g_{k+1,k+1} = ax\ and\ \ g_{k+1,k}g_{k,k+1} = -b$ Putting these two values we have

$|A(k+1)| = a|A(k)| + b|A(k-1)|$

By our hypothesis result is true for $n \leq k$

Using this hypothesis

$|A(k+1)| = aG_{k+1} + bG_k = G_{k+2}$

So result is true for $n = k + 1$

Which proves the result is true $for\ all\ n$

Corollary 1

Consider a sequence of matrix particular case of (6) by putting, $\quad a = 1, b = 1\ in\ (6)$

Now we will prove this result for Fibonacci sequence of polynomials

$$[g_{i,j}] = \begin{Bmatrix} g_{i,j} = x \quad if\ \ i = j \\ g_{i,j= -1} \quad if\ i = j + 1 \\ g_{i,j} = 1\ \ if\ \ i = j - 1 \\ g_{i,j} = 0\ \ if\ oterwise \end{Bmatrix} \qquad (10)$$

$$\boldsymbol{C(n)} = \begin{bmatrix} \boldsymbol{x} & \boldsymbol{-1} & \boldsymbol{0} & \dots & \dots & \boldsymbol{0} \\ \boldsymbol{1} & \boldsymbol{x} & \boldsymbol{-1} & \dots & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{1} & \boldsymbol{x} & \cdots & \dots & \boldsymbol{0} \\ \boldsymbol{0} & \cdots & & \cdots & \dots & \boldsymbol{0} \\ \cdots & \dots & \cdots & \cdots & \cdots & \boldsymbol{-1} \\ \boldsymbol{0} & \cdots & \cdots & \cdots & \boldsymbol{1} & \boldsymbol{x} \end{bmatrix}$$

Then determinant of $\boldsymbol{C(n)}$ are of the type

$$|C(1)| = g_{1,1}$$

$$|C(2)| = g_{1,1}g_{2,2} - g_{2,1}g_{1,2}$$

$$\cdots$$

$$|C(n)| = g_{n,n}|C(n-1)| - g_{n,n-1}g_{n-1,n}|C(n-2)| \tag{11}$$

Statement: - $|C(n)| = F_{n+1}(x)\ for\ all\ n \geq 1$, where $|C(n)|$ is the determinant of nth term of above define (11) sequence of matrix and $F_{n+1}(x)$ is $(n+1)th$ term Fibonacci sequence of polynomials.

Proof: - We will prove this result by principle mathematical induction

For $n = 1$ we have $|C(1)| = x$ also $F_2(x) = x$

So, we can say $|C(1)| = F_2(x)$

So result is true for $n = 1$

Consider result is true for $n \leq k$ (our hypothesis)

So we have $|C(n)| = F_{n+1}(x) \quad for\ all\ \ n \leq k$

Now we will show that result is also true for $n = k + 1$

Consider $|C(k+1)| = g_{k+1,k+1}|C(k)| - g_{k+1,k}g_{k,k+1}|C(k-1)| \qquad by\ (11)$

$By\ (10)\quad g_{k+1,k+1} = x\ and\ \ g_{k+1,k}g_{k,k+1} = -1$ Putting these two values we have

$|C(k+1)| = x|C(k)| + |C(k-1)|$

By our hypothesis result is true for $n \leq k$

Using this hypothesis we have

$|C(k+1)| = xF_{n+1}(x) + F_n(x) = F_{n+2}(x)$

So result is true for $n = k + 1$

Which proves the result is true $for\ all\ n$

Theorem 2: - $|D(n)| = V_{k+1}\ for\ all\ n \geq 1$, where $|D(n)|$the determinant of nth term of above is define (8) sequence of matrix and V_{k+1} is $(k + 1)th$ term a particular case of generalized Fibonacci sequence (4)

Proof: - We will prove this result by principle mathematical induction

For $n = 1$ we have $|D(1)| = a$ also $V_2 = a$

So, we can say $|D(1)| = V_2$

So result is true for $n = 1$

Consider result is true for $n \leq k$ (our hypothesis)

So we have $|D(n)| = V_{k+1}$

Now we will show that result is also true for $n = k + 1$

Consider $|D(k + 1)| = g_{k+1,k+1}|D(k)| - g_{k+1,k}g_{k,k+1}|D(k - 1)|$

By (8) $g_{k+1,k+1} = a\ and\ \ g_{k+1,k}g_{k,k+1} = -b$ Putting these two values we have

$|D(k + 1)| = a|D(k)| + b|D(k - 1)|$

By our hypothesis result is true for $n \leq k$

Using this hypothesis we have

$|D(k + 1)| = aV_{k+1} + bV_k = V_{k+2}$

So result is true for $n = k + 1$

Which proves the result is true $for\ all\ n$

Corollary 2

Consider a sequence of matrix particular case of (1.7) by putting, $a = 1, b = 1$ in (1.7)

Now we will prove this result for Fibonacci sequence of numbers

$$[g_{i,j}] = \left\{ \begin{array}{c} g_{i,j} = 1 \quad if\ \ i = j \\ g_{i,j= -1} \quad if\ i = j + 1 \\ g_{i,j} = 1\ \ if\ \ i = j - 1 \\ g_{i,j} = 0\ \ if\ oterwise \end{array} \right\} \tag{12}$$

$$\boldsymbol{T(n)} = \begin{bmatrix} \mathbf{1} & \mathbf{-1} & \mathbf{0} & \dots & \dots & & \mathbf{0} \\ \mathbf{1} & \mathbf{1} & \mathbf{-1} & \dots & \cdots & & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \cdots & \dots & & \mathbf{0} \\ \mathbf{0} & & \cdots & & \cdots & \dots & \mathbf{0} \\ \cdots & \dots & & \cdots & \cdots & \cdots & \mathbf{-1} \\ \mathbf{0} & & \cdots & \cdots & \cdots & \mathbf{1} & \mathbf{1} \end{bmatrix}$$

Then determinant of $\boldsymbol{T(n)}$ are of the type

$$|T(1)| = g_{1,1}$$

$$|T(2)| = g_{1,1}g_{2,2} - g_{2,1}g_{1,2}$$

$$\cdots$$

$$|T(n)| = g_{n,n}|T(n-1)| - g_{n,n-1}g_{n-1,n}|T(n-2)| \tag{13}$$

Statement: - $|T(n)| = F_{n+1}\ for\ all\ n \geq 1$, where $|T(n)|$ is the determinant of nth term of above define (12) sequence of matrix and F_{n+1} is $(n+1)th$ term Fibonacci sequence of number.

Proof: - We will prove this result by principle mathematical induction

For $n = 1$ we have $|T(1)| = 1$ also $F_2(x) = 1$

So we can say $|T(1)| = F_2$

So result is true for $n = 1$

Consider result is true for $n \leq k$ (our hypothesis)

So we have $|T(n)| = F_{n+1}\quad for\ all\ n \leq k$

Now we will show that result is also true for $n = k + 1$

Consider $|T(k+1)| = g_{k+1,k+1}|T(k)| - g_{k+1,k}g_{k,k+1}|T(k-1)|$ $\quad by\ (13)$

$By\ (12)$ $\quad g_{k+1,k+1} = 1\ and\ \ g_{k+1,k}g_{k,k+1} = -1$ Putting these two values we have

$|T(k+1)| = |T(k)| + |T(k-1)|$

By our hypothesis result is true for $n \leq k$

Using this hypothesis we have

$|T(k+1)| = F_{n+1} + F_n = F_{n+2}$

So result is true for $n = k + 1$

Which proves the result is true $for\ all\ n$

References

[1] Pavel Trojovsk´y, ON A SEQUENCE OF TRIDIAGONAL MATRICES WHOSE DETERMINANTS ARE FIBONACCI NUMBERS Fn+1, International Journal of Pure and Applied Mathematics Volume 102 No. 3 2015, 527-532

[2] N. D. Cahill, D. A. Narayan, Fibonacci and Lucas numbers as tridiagonal matrix determinants. Fib. Quart. 42, No. 3, (2004), 216–221.

[3] K. Kaygisiz, A. S¸ahin, Determinant and Permanent of Hessenberg Matrix and Fibonacci Type Numbers, Gen. Math. Notes 9, No. 2 (2012), 32–41.

[4] E. Kili¸c, D. Ta¸sci, Negatively subscripted Fibonacci and Lucas numbers and their complex factorizations, Ars Combin. 96 (2010), 275–288.532 P. Trojovsk´y

[5] T. Koshy, Fibonacci and Lucas numbers with applications, John Wiley & Sons, 2011.

[6] G. Y. Lee, S. G. Lee, A note on generalized Fibonacci numbers, Fib. Quart., 33, No. 3 (1995), 273–278. [7] G. Y. Lee, J. S. Kim, The linear algebra of the k-Fibonacci matrix, Linear Algebra Appl. 373 (2003), 75–87.

[8] A. Nalli, H. Civciv, A generalization of tridiagonal matrix determinants, Fibonacci and Lucas numbers, Chaos Solitons Fractals 40, No. 1 (2009), 355–361.

[9] A. A. Ocal, N. Tuglu, and E. Altini¸sik, On the representation of k ˝ - generalized Fibonacci and Lucas numbers, Appl. Math. Comput. 170 (2005), 584–596.

[10] G. Strang, Linear algebra and its applications, Brooks/Cole, 3rd edition, 1988.

[11] F. Yılmaz, T. Sogabe, A note on symetric k-tridiagonal matrix family and the Fibonacci numbers, Int. J. Pure and Appl. Math., 96, No. 2 (2014),

[12] N. D. Cahil, J. R. D'Errico, and J. P. Spence, Complex Factorizations of the Fibonacci and Lucas Numbers, Fib. Quart. 41, No. 1 (2003), 13–19.